

Algebraic Manipulation

Algebraic manipulation involves applying algebraic operations, such as addition, subtraction, multiplication, and division, to reorganize equations. This process often aims to isolate a particular variable, making it easier to solve equations or understand the relationship between different variables.

Examples

1. Addition: Make d the subject of the expression $b = c - d$.

To make d the subject of the expression $b = c - d$, is necessary to isolate d on one side of the equation. Here's how to do it:

Given: $b = c - d$

Add d to both sides to isolate d :

$$b + d = c - d + d$$

$$b + d = c - \cancel{d} + \cancel{d}$$

$$b + d = c$$

Now, subtract b from both sides to isolate d :

$$b + d - b = c - b$$

$$\cancel{b} + d - \cancel{b} = c - b$$

$$d = c - b$$

So, d is the subject of the expression $b = c - d$, and it can be written as $d = c - b$.

2. Subtraction: Make r the subject of the expression $p = q + r$.

To isolate r in the expression $p = q + r$, you can rearrange the equation by subtracting q from both sides. Here's how to make r the subject:

Given: $p = q + r$

Subtract q from both sides:

$$p - q = q + r - q$$

$$p - q = \cancel{q} + r - \cancel{q}$$

$$r = p - q$$

Thus, the expression that makes r the subject is $r = p - q$.

3. Multiplication: Make d the subject of the expression $p = \frac{\pi d}{2}$.

To make d the subject of the expression $p = \frac{\pi d}{2}$, is necessary to isolate d on one side of the equation. Here's how to do it:

Given: $p = \frac{\pi d}{2}$

Multiply both sides by $\frac{2}{\pi}$ to isolate d :

$$\frac{2}{\pi} \cdot p = \frac{2}{\pi} \cdot \frac{\pi d}{2}$$

$$\frac{2}{\pi} \cdot p = \frac{\cancel{2}}{\cancel{\pi}} \cdot \frac{\pi \cancel{d}}{\cancel{2}}$$

$$\frac{2p}{\pi} = d$$

So, d is the subject of the expression $p = \frac{\pi d}{2}$, and it can be written as $d = \frac{2p}{\pi}$.

4. Division: Make a the subject of the expression $mz = ad$.

To make a the subject of the expression $mz = ad$, is necessary to isolate a on one side of the equation. Here's how to do it:

Given: $mz = ad$

Divide both sides by d to isolate a :

$$\frac{mz}{d} = \frac{ad}{d} \Rightarrow \frac{mz}{d} = \frac{a\cancel{d}}{\cancel{d}}$$

$$\frac{mz}{d} = a$$

So, a is the subject of the expression $mz = ad$, and it can be written as $a = \frac{mz}{d}$.